




# THE NATIONAL INSTITUTE OF ENGINEERING

*(An Autonomous Institute under VTU, Belagavi)*

**MYSURU - 570 008**

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

*(Artificial Intelligence & Machine Learning)*

---

# Banknote Authentication Using Logistic Regression

*A Machine Learning Laboratory Project Report*

---

In partial fulfillment of the requirements for the award of the degree of

**Bachelor of Engineering**

in

**Computer Science and Engineering (Artificial Intelligence & Machine Learning)**

## Submitted by:


| | |
|---|---|
| **Name of Student** | Shreyas J |
| **USN** | 4NI24CI104 |
| **Semester & Section** | 5th Semester, Section B |
| **Lab Batch** | F2 |


**Academic Year: 2026 – 2027**

# Contents

**1 Introduction and Problem Statement** **2**
1.1 Problem Statement . . . . . 2
1.2 Project Workflow Architecture . . . . . 2

**2 Dataset Information & Exploratory Data Analysis** **2**
2.1 Dataset Features . . . . . 2
2.2 Exploratory Analysis Observations . . . . . 3

**3 Methodology & Preprocessing** **3**
3.1 Dataset Splitting . . . . . 3
3.2 Feature Scaling & Prevention of Data Leakage . . . . . 3

**4 Model Training & Predictions** **3**
4.1 The Sigmoid Function . . . . . 3

**5 Evaluation Metrics & Confusion Matrix** **5**
5.1 Performance Metrics . . . . . 5
5.2 Confusion Matrix Interpretation . . . . . 5

**6 Threshold Investigation** **6**

**7 Detailed Answers to Lab Questions** **6**

**8 Conclusion** **7**

# 1 Introduction and Problem Statement

Machine learning classification plays a fundamental role in automated pattern recognition and financial security systems. Unlike regression tasks where the target output is continuous, binary classification focuses on assigning records into one of two discrete categories. This project investigates the application of **Logistic Regression** to classify scanned banknote images into authentic or counterfeit categories using statistical features derived from wavelet-transformed images.

## 1.1 Problem Statement

A banking institution requires an automated screening pipeline to process quantitative image measurements extracted from banknotes. The objective is to construct, train, evaluate, and interpret a binary Logistic Regression classifier operating on four numerical features.

> *Important Note: This project is an educational exercise utilizing a public benchmark dataset. The model is not certified for deployment in live banking environments, commercial currency verification devices, or legal enforcement applications.*

## 1.2 Project Workflow Architecture

The overall end-to-end Machine Learning pipeline is depicted in Figure 1.

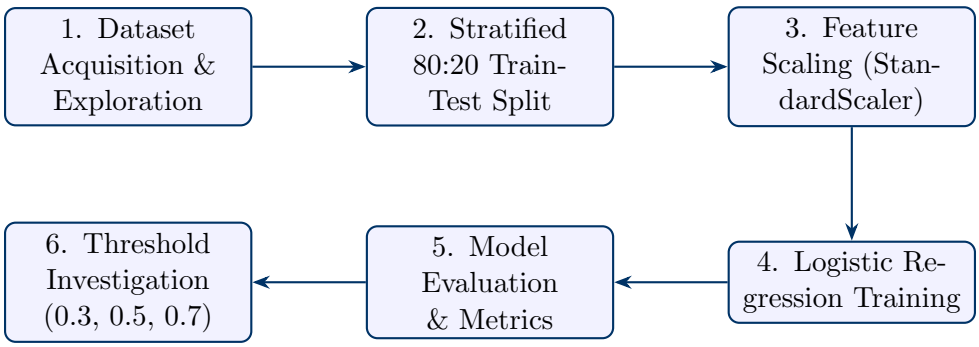



Figure 1: End-to-End Machine Learning Project Workflow.

# 2 Dataset Information & Exploratory Data Analysis

The **Banknote Authentication Dataset** was sourced from the UCI Machine Learning Repository. It comprises 1,372 records derived from continuous wavelet transform (CWT) features of digitized banknote images.

## 2.1 Dataset Features

- **Variance:** Variance of the wavelet-transformed image (Continuous numerical input).
- **Skewness:** Skewness of the wavelet-transformed image (Continuous numerical input).
- **Curtosis:** Kurtosis measurement of the wavelet-transformed image (Continuous numerical input).
- **Entropy:** Entropy of the image (Continuous numerical input).
- **Class:** Binary target variable (0 or 1). According to official UCI documentation, Class 0 represents genuine banknotes and Class 1 represents forged banknotes.

### 2.2 Exploratory Analysis Observations

Exploratory data analysis revealed zero missing values and zero duplicate records across all 1,372 instances. The target variable comprises 762 genuine notes (55.5%) and 610 forged notes (44.5%), confirming a balanced binary dataset.

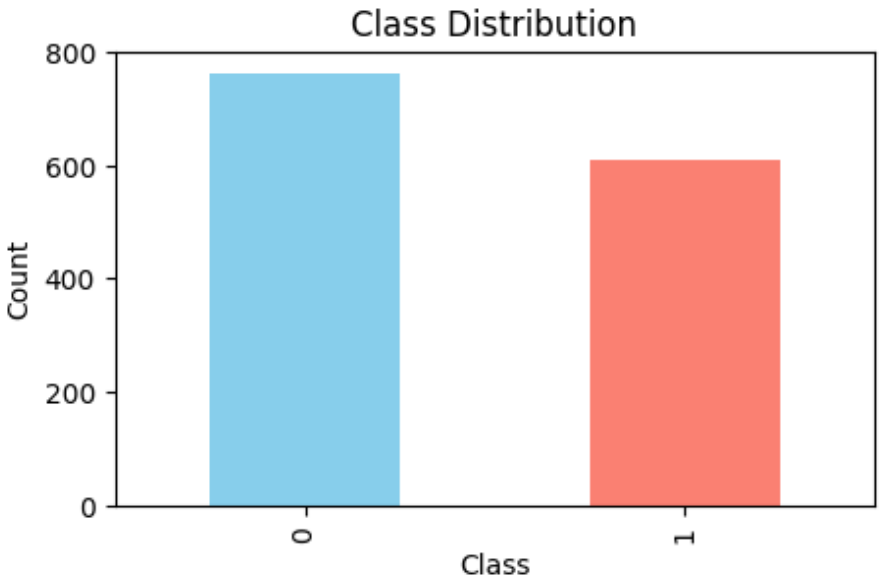



Figure 2: Class distribution of genuine (Class 0) vs. forged (Class 1) banknotes.

## 3 Methodology & Preprocessing

### 3.1 Dataset Splitting

The dataset was partitioned into 80% training data (1,097 samples) and 20% test data (275 samples). Stratified sampling was enforced with `random_state=42` to maintain equal target class distribution across subsets.

### 3.2 Feature Scaling & Prevention of Data Leakage

Because features exhibit varying scale distributions, `StandardScaler` was used. To avoid **data leakage**, parameters $(\mu, \sigma)$ were calculated **only** on the training subset ($X_{train}$), and applied to transform both $X_{train}$ and $X_{test}$.

## 4 Model Training & Predictions

The classifier was initialized using Scikit-learn's `LogisticRegression(max_iter=1000)` to ensure solver convergence.

### 4.1 The Sigmoid Function

The mathematical model projects a linear combination of input features into a probability range $[0, 1]$ using the sigmoid function:

$$P(y = 1|\mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad \text{where } z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 \tag{1}$$

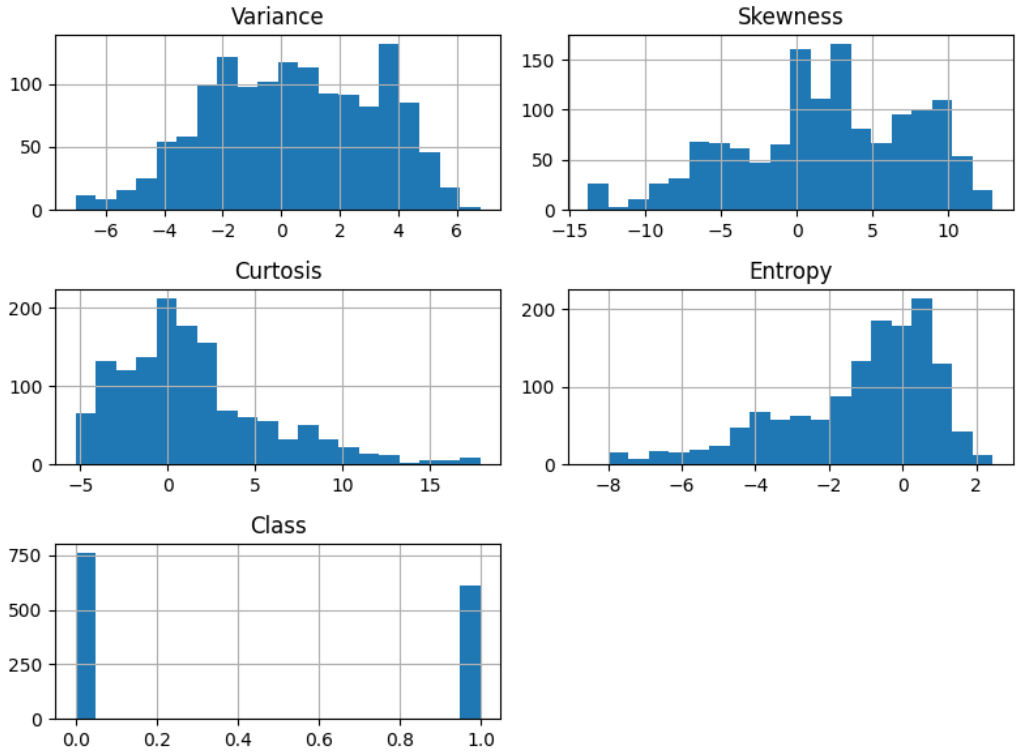



Figure 3: Histograms displaying input feature distributions.

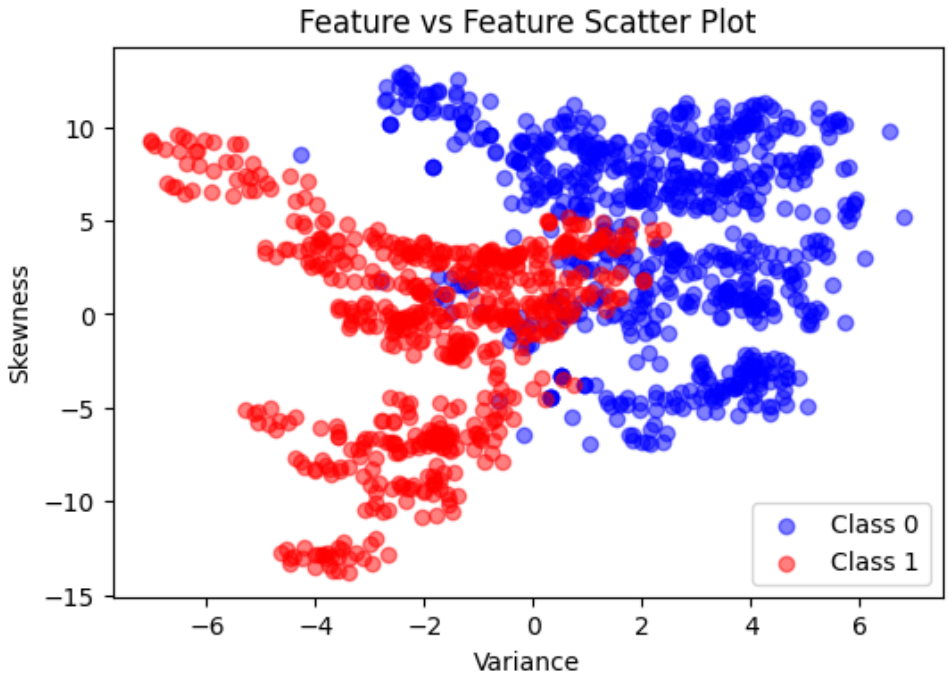



Figure 4: Feature-versus-feature scatter plot illustrating visual class separability.

# 5 Evaluation Metrics & Confusion Matrix

## 5.1 Performance Metrics

Evaluation on the 275 unseen test instances produced the following performance metrics:

| Metric | Observed Value |
|---|---|
| Accuracy | 98.55% |
| Precision (Class 1) | 97.30% |
| Recall (Class 1) | 99.10% |
| F1-Score (Class 1) | 98.19% |

Table 1: Summary evaluation metrics on the test dataset.

## 5.2 Confusion Matrix Interpretation

The test set confusion matrix layout is $\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$.

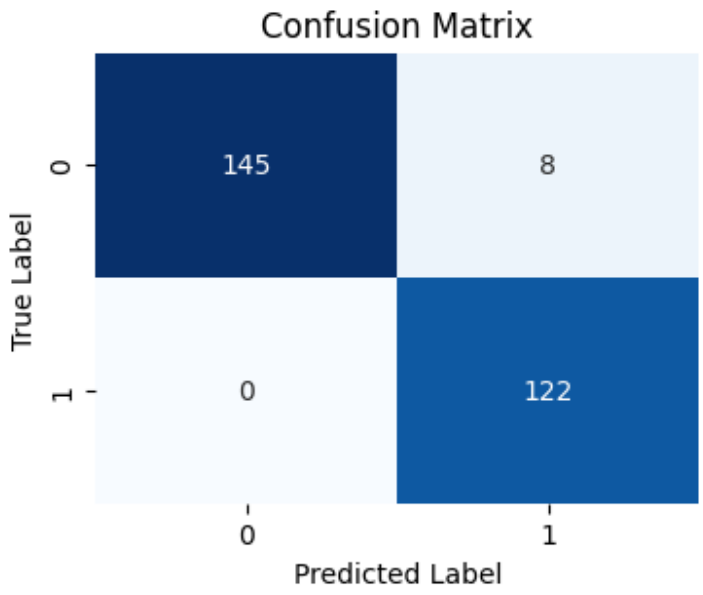



Figure 5: Confusion matrix visual representation.

# 6 Threshold Investigation

The default threshold for converting predicted probability to class label is 0.50. Modifying this decision boundary alters classification performance:

| Threshold | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| 0.30 | 97.82% | 94.12% | 100.00% | 96.97% |
| 0.50 | 98.55% | 97.30% | 99.10% | 98.19% |
| 0.70 | 97.45% | 99.07% | 94.59% | 96.78% |

Table 2: Comparative performance across decision thresholds (0.30, 0.50, 0.70).

# 7 Detailed Answers to Lab Questions

1. **What is the difference between regression and classification?**
   Regression models predict continuous numerical outputs (e.g., house prices), whereas classification models predict discrete category labels (e.g., genuine vs. counterfeit).

2. **Why is Logistic Regression used for classification even though its name contains the word regression?**
   It uses an underlying linear regression equation to calculate log-odds of a class, which is then mapped into a probability bounded between 0 and 1 via the sigmoid function.

3. **What are the four input features in the Banknote Authentication dataset?**
   The four inputs are Variance, Skewness, Curtosis, and Entropy derived from wavelet-transformed image scans.

4. **What is the target variable? What do Class 0 and Class 1 represent?**
   The target variable is 'Class'. Class 0 represents genuine banknotes and Class 1 represents forged banknotes.

5. **How many records and features are present? Are there missing or duplicate records?**
   There are 1,372 records with 4 input features and 1 target column. There are zero missing values and zero duplicate records.

6. **Is the class distribution balanced?**
   Yes. Class 0 comprises 55.5% (762 records) and Class 1 comprises 44.5% (610 records), indicating a well-balanced distribution.

7. **Why do we divide the dataset into training and testing subsets?**
   To evaluate model generalization performance on unseen records and prevent overfitting.

8. **What is the purpose of stratified splitting?**
   Stratified splitting ensures that proportional class representation is preserved identically across training and testing splits.

9. **Why is feature scaling useful for Logistic Regression?**
   Features with wide numerical ranges can dominate optimization; scaling accelerates gradient convergence and ensures equal weight contribution.

10. **What is data leakage? How can fitting a scaler before the train-test split cause it?**

Data leakage occurs when test set statistical information contaminates model training. Fitting a scaler before splitting exposes test mean and variance to the training pipeline.

11. **What does the sigmoid function do in Logistic Regression?**
It maps any real-valued number into a value between 0 and 1, converting log-odds to class probabilities.

12. **What does a predicted probability of 0.80 for Class 1 mean?**
It means the model estimates an 80% probability that the banknote is counterfeit (Class 1).

13. **Explain accuracy, precision, recall, and F1-score in your own words.**

- *Accuracy:* Overall fraction of correct predictions out of all predictions.
- *Precision:* Fraction of correctly identified forged notes out of all notes labeled as forged.
- *Recall:* Fraction of actual forged notes correctly detected by the model.
- *F1-score:* Harmonic mean balancing precision and recall.

14. **Explain TP, TN, FP, and FN in the context of this project.**

- *True Positive (TP):* Forged note correctly predicted as forged.
- *True Negative (TN):* Genuine note correctly predicted as genuine.
- *False Positive (FP):* Genuine note incorrectly predicted as forged.
- *False Negative (FN):* Forged note incorrectly predicted as genuine.

15. **How did metrics change when the threshold changed from 0.30 to 0.50 and 0.70?**
Lowering threshold to 0.30 increased Recall to 100% (zero False Negatives) but lowered Precision. Raising threshold to 0.70 increased Precision to 99.07% but reduced Recall.

16. **Which metric would you prioritize if failing to identify a forged note were especially costly?**
**Recall** must be prioritized to minimize False Negatives, ensuring counterfeit notes do not enter financial circulation unnoticed.

17. **State at least two limitations of this dataset or your model.**

(a) Features are computed from legacy grayscale wavelet transformations rather than high-resolution color security scans.

(b) Linear decision boundaries of Logistic Regression cannot capture complex non-linear feature interactions without feature engineering.

# 8 Conclusion

The Logistic Regression model achieved 98.55% accuracy on unseen test data. Lowering the classification threshold to 0.30 achieved 100% recall, successfully eliminating False Negatives—the primary risk in banknote validation pipelines. Future deployment requires continuous hardware calibration, non-linear baseline evaluations, and compliance with institutional financial standards.